

WHestBench Phase-1 Algorithmic Contribution (v13, corrected) - #326094

Phase-1 Algorithmic Contribution writeup — draft v12

Status: DRAFT v13, 2026-08-17. **Section landing and citation pinning.** v13 replaces §1 with the finished estimator-and-constants section (the selected-constant count is **seven** — `on_alpha = 3.0` was omitted by every draft before 2026-08-12), inserts the design-axis closure as §3b2, corrects the repository-visibility statement in §6, and pins the reproducibility citation to a pushed commit.

v12, 2026-08-11: a correction release, and a restructure. v10–v11 corrected five defects we found in our own v9 after filing it; v12 reorganises the document around what the algorithmic-contribution guidelines actually judge — understanding of white-box estimation, mechanistic content, and full transparency about LLM involvement — rather than around a score.

The five errata are stated in full in §0 and marked inline at each site (E1 §3 table and §3f, E2 §4, E3 §3d, **E4 §3e/§3f, E5 §3e(5)**). E1–E3 and E5 change published numbers and one recommendation. **E4 changes a conclusion:** the central synthesis was stated as a theorem, it is not one, and its corrected status is open. We would rather file a weaker paper that is true than a stronger one that is not, and we would rather say so on the first page than in a footnote — a document whose §4 tells other people not to carry dispositions by wording cannot carry its own claims that way.

v9, 2026-08-10: adversarial-closure pass — new §3f (the Gen-7 twenty-agent campaign, the seed-side SVD-V null, the 7/7 re-litigated kill families, S18's singleton-cell seal) and a falsification-hygiene paragraph in §4; v8's level rules carry over unchanged, and the floor remains a lower-bound attempt. v8: level-repair pass — §5 adjudication removed, floor language set to S17's earned framing, §3c/3d dispersion corrected per S1B, residual restated. **Graded submission ID: #326094** (adjusted 1.832e-7, 50/50 public MLPs, 0 failures, rank #58 at grading time). Filing deadline Aug 17, 23:59 UTC per the Algorithmic Contribution guidelines thread.

Beyond Gaussian closure: structured spherical quadrature for deep random

ReLU networks — and a map of which white-box improvement families fail, why,

and what remains open

Executive summary

What this contributes is a map of the boundary, not a score. The estimator we submit is a vehicle for the measurements; the durable content is a set of statements about what no method of a given shape can achieve on these networks, each stated at exactly the confidence its evidence earns — including four claims we withdrew during the week, with their arithmetic.

The mechanism, and an exact map of which parts are forced and which are chosen. The estimator integrates over a frozen phased-Hadamard **exact spherical 2-design** (126 mutually unbiased frames $\times$ 256 directions, antipodally doubled to 64,512) at the exact radius $E\|X\|$, with structural pruning, terminal-layer folding, and a first-layer moment-tangent control.

All values below are read from the **deployed** class, `kerdock_v3_estimator.py`, not from the base classes it inherits from — a distinction that cost us two wrong statements before we got it right (see the correction note).

Forced by the construction, with nothing tunable in them: the design itself; the sample count $n_{\text{base}} = 126 \times 256 = 32,256$ base directions, antipodally doubled to 64,512 — this is the design’s size, not a budget anyone picked; the exact radius $E\|X\| = 15.98438266660852747\dots$ from the chi moments, applied via `radial_conditioning = True`; and the uniform weights, which are not a convenience but provably the constrained optimum at every degree (§3b, and the proof in companion P4).

Selected during development, and therefore fitted in the sense that matters to an auditor: the moment-tangent coefficient $\lambda = 0.9807112198896164$ (inherited and active), the pilot sizes `pilot_base = 256` and `fold_pilot_base = 1,024`, the fold thresholds `dead_alpha = -2.0` (inherited and active) and `on_alpha = 3.0` (live in the terminal fold; omitted by every draft before 2026-08-12 — §1 carries the disclosure), and the phase window `phase_start = 2`, `phase_stop = 128`. Seven constants, all scalar, all frozen before grading.

Correction, stated here because this very paragraph got it wrong twice. A prior v12 draft claimed “every constant is forced,” “none are tuned,” and “zero fitted structure anywhere in the estimator.” **False** — the seven constants above are selected. An adversarial audit caught it within the hour. The repair was then *also* wrong: it listed $n_{\text{base}} = 14,000$ and a decision to disable radial conditioning as fitted choices, both read from the base class, when the deployed subclass overrides them to 32,256 and `True`. A second audit caught that within fifteen minutes.

We leave both errors on the page rather than quietly landing the third version, because the failure mode is exactly the one §4b is about — asserting a property of a system from a proxy rather than from the artifact that ships — and because a paper claiming an evidence discipline should show the discipline failing and being caught, not only succeeding.

What survives, precisely: the N8c screen detects no material final-layer bias (point estimate -0.0336 against a 0.25 kill threshold, three nets and sixteen rotations — an observation, not a theorem; see E6), and no component fits to the *evaluation* suite. **Near-zero measured bias does not prove absence of fitting**, and we no longer claim it does. The defensible statement is that the fitted parts are few, enumerated above, frozen before grading, and confined to scalar budget and correction coefficients rather than to anything that could learn the target.

Four boundary results, each stated with its exact scope. Three are proved; the fourth is a measured screen and is labelled as one.

1. **Fixed, output-independent reweighting of a fixed design cannot help, under a zonal Haar-averaged criterion.** [proved] Positive-semidefiniteness plus constant row sums makes uniform weighting a global minimiser of the quadrature error at *every* spherical-harmonic degree, and for every nonnegative mixture of degrees. **Scope, and it is narrow on purpose:** this

closes *fixed, output-independent* weights against a *zonal* criterion. It does **not** close arbitrary reweighting — not weights that depend on the realised network, not adaptive rules (which would need their own unbiasedness proof), not changes to the point set, and not non-zonal criteria such as the realised output covariance. **Uniqueness is weaker still, and weakest on the set we actually deploy.** On the 32,256-point base set, uniform is the *unique* minimiser only where the kernel is positive definite — verified at degrees 4, 6 and 8; **false at degree 2**, where the design is exact and every per-frame reweighting is free; open above. On the **antipodally doubled 64,512-point set the estimator actually uses**, uniqueness fails at *every* even degree: the even-degree kernel has the block form $\mathbb{I}_2 \otimes \kappa$, whose kernel contains every antipodally antisymmetric perturbation, so the minimum is attained on a 32,256-dimensional affine set containing uniform. **Uniform is therefore a minimiser of our deployed design, never the unique one.** That does not weaken the closure — a global minimiser is all the closure needs — but “the constrained minimiser” would be the wrong phrase and we no longer use it.

2. **A whole family of truth-free estimators is dead by algebra.** If an estimator anchors on its own uniform frame mean, the sum-one GLS solution is uniform *identically* — for every ridge and every shrinkage, with no probabilistic model. Four descendants that failed in four apparently different ways are one failure, and the identity predicts it without an experiment.
3. **Within one admissible class of surface rewrites, there are exactly two kinds and no third.** [proved, under stated hypotheses] Either the rewrite is free of the kink set — in which case it collapses to reweighted point evaluations whose entire radial content is Euler’s identity — or it deposits mass on the kink set, and realizing *that rewrite* means locating kinks and evaluating gradient jumps. **Scope, and it is the narrowest of the four:** exhaustiveness holds over a specific admissible class (fields jointly locally Lipschitz in all three slots, with an $o(|x|^{1-d})$ bound at the origin) and **fails without those hypotheses** — a third class exists under weaker ones, and we exhibit it. The result classifies **representations of the integral**, so “requires locating K” describes what a rewrite of this shape *contains*; it is **not an algorithmic lower bound** on estimating the target by other means. Companion P5 carries the proof, the counterexample, and the open items.
4. **An exact surface identity for the Gaussian mean exists, and the estimator it induces is far worse than sampling on the screen we ran.** [measured, not proved] $E[f(X)]$ equals a Gaussian-weighted integral of normal gradient jumps over the kink set — that identity is exact and independently implemented twice. The induced estimator lost to Monte Carlo by $\sim 189\times$ in variance (per-seed 196.0 / 173.3 / 199.9) even granting a free oracle for every crossing and jump. **Scope:** that is a measurement at the widths and depths screened, not a theorem about every kink-set method.

What we do NOT claim, and withdrew during the week: that the residual is maximum-entropy; that its harmonic spectrum is characterized; that any truncation class is closed; or that a minimax floor is established. Those are **open**, and §0 gives the four errata with their evidence.

The falsification protocol is the second contribution, and §4b reports it being turned on ourselves under adversarial conditions.

o. Errata against the v9 filed 2026-08-10

Five defects in the filed text, all found by our own post-filing audit. Three change a published number, one retracts advice we gave the field, and **one changes a conclusion**. All five are stated here rather

than patched quietly, and each is also marked at its site.

E1 — the 0.00% f64-lane figure came from a structurally void detector (§3 ledger table, §3f). `run_m183_falsifier.py:58` reads `dtypes = getattr(op, "dtypes", None) or ()`, and the installed flopscope 0.10.0 `OpRecord` has no `dtypes` field — its fields are `count`, `cumulative`, `flop_cost`, `flopscope_backend_duration_s`, `flopscope_context_start_offset_s`, `flopscope_overhead_duration_s`, `index`, `namespace`, `op_name`, `resolved_dtype`, `shapes`, `subscripts`. The guard therefore evaluated `any(...)` over an empty tuple for every op on every program, and 0.00% was the only output it could ever produce. Two independent signals establish this: the dataclass field list read from the pinned `venv`, and the detector run against a deliberately 100%-float64 program (five 64x64 f64 matmuls) reporting `f64_share 0.0` while a corrected `resolved_dtype` detector reports 1.0 on the same log. The honest measurement is **1.193e8 FLOPs, 0.0755% of predict, with a full-recast ceiling of 59,656,312 FLOPs** — which independently reproduces the 59.66M our Gen-7 cost-remap attacker derived by a different route. The M183 kill stands, and the dtype-repricing escape stays closed, because the corrected ceiling is still immaterial. What does not stand is the number.

E2 — the 1.65 suite-calibration ratio is a mean artifact, and we recommended it (§4).

Recomputing the same committed 22-net panel by median gives 6.47355e-7 against the grader's published 6.470e-7, a 0.05% match. The 1.65 came entirely from the right tail of a per-net distribution with roughly an 11x spread. There is no suite-difficulty shift, our filed advice to apply that factor was wrong, and §4 now carries the corrected recommendation: calibrate on the median.

E3 — the fresh-seed band treated its own anchor as exact (§3d). The 1.830e-7 anchor is itself a 50-net measurement carrying a 9.83% standard error. Folding that in widens the honest 50-net P5-P95 band from [1.54e-7, 2.16e-7] to **[1.46e-7, 2.25e-7]** and raises $P(\text{suite score} > 2.5e-7)$ from 0.034% to **0.57%**. We state the wider band.

E4 — the central synthesis was stated as a theorem and is not one (§3e, §3f). This is the largest correction in the document and it goes against our own headline. v9 asserted that the finite-width output of a deep random ReLU network is *maximum-entropy* independent χ^2_1 speckle sitting at the degree-4 boundary of an exact 2-design, and that this one fact explains every failed mutation, the champion's correction-proofness, and a floor set by an independent-cell count N_{eff} . **Withdrawn.** Three separate defects:

1. *The evidence is quarantined.* The harmonic-spectrum computation that appeared to establish the mechanism was executed after a sealed evidence charter and was not disclosed as in flight, so it carries no evidence weight. Its outputs are retained and visibly marked rather than deleted, because deleting an inadmissible measurement is worse than quarantining it.
2. *The older account does not revive.* Withdrawing the replacement does not restore equipartition — the record never established that either. The degree- ℓ design-error operator and the residual field's harmonic energy coefficients are different objects, and flatness of the former implies nothing about the latter. A χ^2 -like one-point amplitude marginal does not determine a harmonic spectrum.
3. *Family-local results were promoted to global ones.* Output-side feature families failing at the frozen target is a strong family-local observation. It is not a spectral-tail theorem, a no-truncation theorem, or a minimax floor, and v9 used it as all three.

Corrected status: **open**, not reverted. What survives is stated in §3e in the narrower form it earns, and it is still enough for the claim that matters — the champion’s final-layer bias is measured at zero (N8c) and no component of it fits the evaluation suite. That is a measurement and it needs no entropy argument. It is **not** a claim that the estimator contains no fitted constants — it does, and they are enumerated in the executive summary.

Two audits reached this independently and within hours of each other: an adversarial partner auditing our papers under a sealed evidence protocol, and our own twelve-agent refinement pass, which separately flagged that the correction had been inserted into one section and never propagated to the four that depend on it. Neither audit was prompted by the other. We record that because it is the strongest evidence we can offer that the correction is real rather than defensive.

A note on how these were found, since it bears on §4’s argument. E1 and E2 both surfaced from an adversarial pass over our own record run against public forum material — external attack on this work produced no defect, and internal audit produced these three. We take that asymmetry as the strongest evidence available that the record is honest, and we would rather publish the asymmetry than the tidier document.

1. The estimator, and exactly which of its numbers were chosen

The estimator integrates over a frozen phased-Hadamard **exact spherical 2-design**: 126 mutually unbiased frames of 256 directions each, every frame an orthonormal basis $H_{256} \text{diag}(\phi_i)_s/16$, antipodally doubled to 64,512 points at the exact chi-mean radius, with a per-network Haar rotation as the sole randomization. Five components, each independently ablated in §3:

1. **The design**, worth 2.02x isolated against radially conditioned Monte Carlo.
2. **Exact radial conditioning**. A bias-free ReLU network is positively one-homogeneous, so $E[f(X)] = E[|X|] \cdot E[f(U)]$ holds exactly at every layer. The radial degree of freedom is not reduced, it is *removed*.
3. **Pilot-rescued structural pruning**. An analytic diagonal pass marks neurons below a threshold as provisionally cold; a 256-antipodal-pair pilot rescues any that fire. Saves **25.109% of B**.
4. **Three-terminal-layer folding**. Dead columns vanish, always-on columns compose linearly into the next weight matrix, and only kink columns retain a ReLU. Saves **4.828% of B** at an MSE ratio of 1.000033.
5. **A first-layer moment-tangent control**, frozen coefficient, measured neutral on this design.

What is forced, with nothing tunable in it. The design itself; the sample count $n_{\text{base}} = 126 \times 256 = 32,256$, which is the design’s size and not a budget anyone picked; the exact radius $E[|X|] = 15.98438266660852747\dots$ from the chi moments; and the uniform weights, which are *a* global minimiser of the quadrature error at every spherical-harmonic degree under a zonal Haar-averaged criterion — though not the unique one, and on the deployed antipodally doubled set not unique at any even degree.

What was selected during development, and is therefore fitted in the sense an auditor cares about. There are seven.

constant	value	role
moment_tangent_lambda	0.9807112198896164	first-layer control coefficient
pilot_base	256	pilot pairs for the pruning rescue
fold_pilot_base	1,024	pilot pairs for the terminal fold
dead_alpha	-2.0	cold-neuron threshold
on_alpha	3.0	always-on threshold in the fold
phase_start	2	first frame of the deployed slice
phase_stop	128	last frame of the deployed slice

All seven are scalar and all were frozen before grading. `on_alpha` deserves its own sentence, because earlier drafts of this paper enumerated **six** and omitted it: it is the mirror of `dead_alpha`, it is live on the deployed path, and it was swept over $\{3.5, 4.0, 5.0\}$ against `dead_alpha` on development data (ledger record 202) with **all arms flat**. That flatness is worth more than the omission cost — it is direct evidence the estimator is insensitive to a dial we were free to choose, which is the property an auditor is actually looking for.

Two further values are frozen but belong in neither column — the backend block height `BLOCK_ROWS` = 4,096 and the frame ordering — implementation constants carrying no development selection.

We have stated this wrongly twice. An earlier draft claimed “zero fitted structure anywhere in the estimator.” False. The repair was also wrong, reading values from a base class the deployed subclass overrides. Both were caught within the hour by adversarial audit, and both are left on the page rather than quietly replaced, because a paper arguing for an evidence discipline should show that discipline failing and being caught. The count moved from six to seven for the same reason.

What we claim, precisely: the fitted surface is seven scalars, frozen before grading, confined to budget and correction coefficients, containing nothing that could learn the target, and no component was fit to the evaluation suite. Low measured bias does not prove absence of fitting, and we do not claim it does. Every value above was read from the deployed method-resolution order (`estimator.Estimator` → `kerdock_v3` → `fold3` → `base_estimator`) and is re-derived mechanically by `scripts/verify_phase1_writeup.py`.

Measured compute profile, **with the provenance of each number stated, because these are not all from the same machine** (Erratum E9):

- **Local measurements, on our own T4 development host — NOT hosted grader timings:** 2.86 s mean wall per MLP, 4.11 s max against a 60 s cap; residual (the uninstrumented remainder) 0.080 s mean, 0.137 s max. Earlier drafts presented these without provenance, which invited reading them as grader figures. Wall time is hardware-dependent and these do not transfer.
- **Hosted, from the graded ledger for #326094:** adjusted $1.832e-7$ on 50/50 public MLPs with 0 failures, at C/B 0.650.

The residual share — 4.5% of scored C on average, 7.7% of adjusted score in the worst case at C/B 0.650 — is derived from the local timings and therefore inherits their machine dependence. It is conditional on $\lambda = 1e11$ remaining fixed (Rules §5.3 reserves changes), and the organizers have stated they are still

deciding whether Phase 2 keeps residual-time accounting at all. Essentially all arithmetic is instrumented; that claim rests on the FLOP census, not on timing.

2. The non-Gaussianity wall (the main scientific contribution)

Moment/cumulant propagation is the natural white-box approach and the one the organizers' own materials highlight. We built it to the strongest form we could **certify**, then measured exactly how far it falls short.

Certified machinery. A bounded-cost bivariate provider evaluates the rectified-Gaussian pair moment $E[\text{ReLU}(X_i)\text{ReLU}(X_j)]$ and its derivatives via Owen-T/ Φ_2 with per-call enclosure certificates, a fixed 4,048-FLOP budget per pair, charged sign comparisons, exact strata at the rank-one and zero-variance limits, and a 12,890-case adversarial census (all contained). On top of it, an exact zero-order full-covariance recurrence ($\mu_0=0$, $V_0=I$; $a_i = \mu W_i$; $C_i = W_i^T V_{i-1} W_i$; post-ReLU moments from exact Tallis truncated moments) propagates the complete 256×256 covariance through all 32 layers, inclusively metered at $8.30e9$ FLOPs = 3.05% of budget. Assembly agrees with 30-digit mpmath references to $2.144e-9$ and with an independent closure implementation to $\sim 2e-16$.

The measurement. Predicting the depth-32 final layer, against 400k-sample Monte-Carlo truth on He-initialized networks (MC noise floor $1-2e-7$, i.e. 200-1000x below the signal — the numbers are resolved, not noise):

predictor of the depth-32 final-layer mean	bias MSE
diagonal Gaussian closure	$7.18e-4$
exact full-covariance Gaussian closure	$9.61e-5$
sampling estimator (this submission), same budget	$\sim 2.5e-7$

Making the covariance exact buys a factor of ~ 7.5 over the diagonal closure.

The like-for-like gap is 340.9x, raw against raw ($9.6055e-5$ closure against the graded raw $2.818e-7$). That is the only ratio in which numerator and denominator are the same quantity, and it is the one to quote.

Erratum E7. Earlier drafts also printed “ $524\times$ against adjusted $1.832e-7$.” **That figure mixes units** — it divides a *raw* closure MSE by an *adjusted* champion score — and it overstates the gap by an order of magnitude relative to a true adjusted comparison. If the closure were granted the most favourable possible multiplier (the 0.1 floor), adjusted-against-adjusted is $9.6055e-6 / 1.832e-7 = 52.4\times$, not $524\times$. Both figures are stated here so the arithmetic can be checked.

The gap is third-and-higher-cumulant structure that this Gaussian-moment closure does not represent. **Scope, corrected:** earlier drafts said “no Gaussian-moment closure can represent at any compute multiplier.” That is a universal claim and we did not prove it. What we measured is that *this* implementation — a pairwise-exact, assumed-Gaussian recurrence as built — trails by the factor above, and that granting it a zero-cost floor does not close the gap. It kills this implementation as a competitive estimator. It does not prove that no Gaussian-informed method can work.

The design principle this yields: exact Gaussian structure pays when *subtracted* (our moment-tangent control: -19.8% adjusted on its lineage) and fails when *predicted* (closure-as-estimator: 46x

outside the competitive boundary). We think this is the sharpest available statement of why depth-32 white-box estimation is hard, and it is stated with certificates rather than intuition.

3. Falsification ledger — twelve predeclared kills

Every entry was predeclared with its kill gate **before** implementation; none was retuned after seeing results. All are reproducible from the corpus.

mutation	hypothesis	measured	verdict
N4	cheap variance levers	null	killed
N5	multilevel closure control variate	1.07x	killed
N6	exact great-circle Rao-Blackwellization	FoM 0.006x	killed
N7	RQMC superconvergence at depth 32	slopes -0.97/-1.23 vs -1.25 gate	killed
N8a	Kronecker lattice vs our frames	lattice 2.1x worse (CI [1.65,2.65])	killed
N8b	disclosed native backend	0.94e11 FLOP/s < $\lambda = 1e11$	killed
N8c	offline-trained corrector	bias share -0.0336 on a 3-net, 16-rotation screen, against a predeclared kill threshold of 0.25. Erratum E6: the printed CI [-0.031, 0.097] does not contain its own point estimate and is not a usable net-level interval	killed
N9	frames + tangent + deeper fold	+2.1% (positive control +34.5% on iid)	killed
M180	stronger spherical designs (MUB mix / coset rotations / remix)	all arms +20-49% variance	killed
M181	terminal rectified-Gaussian smoothing (3 arms incl. unbiased CV)	bias 4-6x baseline MSE; var identical; $\lambda \rightarrow 0$	killed
M183	float32 hot-path recast (the “free 2x”)	0.0755% f64-lane billing (1.193e8 FLOPs of predict; full-recast ceiling 59,656,312 FLOPs) — no material lane. Erratum E1: v9 filed 0.00% here, from a structurally void detector.	killed

mutation	hypothesis	measured	verdict
M184	mid-layer exact on-composition + sparsity	0.00% billed reduction (certain-on absent where wide; 2.3x under break-even at depth)	killed

M180’s kill carries a structural result: the phased-Hadamard design is **LOCALLY OPTIMAL** — its strength is the mutual unbiasedness of all 126 frames under one shared rotation, and every perturbation (family displacement, rotation fragmentation, per-frame remixing) destroys that negative-covariance structure monotonically. M181’s kill closes the Gaussian-closure family at every insertion point (predictor / control-variate / corrector / smoother), each with an independent measured mechanism of failure — and its adversarial check (the bias field reproduced on fresh iid samples, cosine 0.97-0.98) pins the failure on the terminal law’s non-Gaussianity itself.

Hosted validation of the submission: #326094 graded adjusted 1.832e-7 (final-layer MSE 2.818e-7) on 50/50 public MLPs with zero failures, 3.5x better than the grader’s Monte-Carlo reference, at **5.75 s mean / 6.80 s max** wall per MLP against the 60 s cap (hosted, a1_hosted_ledger.json, 50 rows, min 4.61 s — see E9; earlier drafts printed a local T4 figure here) with a residual exposure of 4.5% of scored C on average (7.7% of adjusted score worst-case at C/B 0.650, conditional on $\lambda = 1e11$ remaining fixed) — the compute profile is essentially fully instrumented, which we note in the context of the accounting discussions elsewhere on this forum.

Two of these are results in their own right. **N8a** established that our spherical design already dominates a randomized lattice — the lattice’s advantage over iid evaporates once the radial degree of freedom is conditioned away. **N8c** established that our estimator’s final-layer error shows **no material final-layer bias on the N8c screen** (point estimate -0.0336 against a 0.25 kill threshold; see E6 — this is an observation on three nets and sixteen rotations, not a zero-bias theorem, and it does not license any inference to the private suite), which is the strongest robustness property available going into a fresh-seed re-evaluation: there is no fitted component to overfit.

Two further proposals die by **exact identity** rather than by statistics, and we record them as identities. The **Crofton kink-transect identity** (experiments/s9_crofton_transect/): the Gaussian mean of a bias-free ReLU network equals the Gaussian-weighted surface integral of its gradient jumps over the kink set — verified at machine precision (structural checks to 6.7e-16, two independent transect estimators, 3 predeclared + 20 fresh nets) — and the unbiased estimator it induces is killed at 176,860x worse variance-per-FLOP than MC on the width-64 depth-8 screen (predeclared kill line 100x). The **residual/norm decomposition identity** (experiments/s16_residual_decomp/, confirmatory; clean statement: “residual decomposition = antipodal symmetrization”): both proposed forms reduce exactly to structure the estimator already has. The even/odd residual split $|z|/2 \pm z/2$ is **BIT-IDENTICAL** to our antipodal pairing — layer-1 identity max deviation exactly 0.0 over 8,257,536 entries on each of 3 nets (IEEE negation is exact and the ReLU pair is a clamp partition), and the full estimators produce bit-equal final outputs (MSE ratio 1.000000, max final deviation 0.0) on all 48 (net, seed) panels — while its “analytic linear + Gaussian corrections” deep reading is the closure family M181 already killed (closure 1.28e-6 vs sampling 3.41e-7): the linear part is exactly zero-mean (its measured mean decays 2.26x when n grows 4x, vs 2.0 predicted for pure noise), so the entire signal lives in the non-Gaussian ReLU corrections. The per-layer residual reparametrization measures $R_l = \text{mean}\|F_l\|/\text{mean}\|y_l\|$ in

[1.108, 1.231] (median 1.162) across all 31 hidden layers — no layer is near-identity, so no perturbative truncation exists and the form reduces to the depth law of §3e(4).

3b. The design is an exact spherical 2-design, and its error spectrum is measured (new)

A deterministic quadrature audit of the 126-frame phased-Hadamard design (64,512 antipodal directions, three independent rotations): odd-degree harmonics are annihilated exactly; degree-2 quadrature error is $8.6\text{e-}9$ — the design is an exact spherical 2-design, which is the harmonic-analysis explanation for its measured local optimality (every perturbation of the family or its shared rotation destroys degree-2 exactness and loses 20-49% variance). The residual angular error lives at degree 4 (11% of the iid level) and degree 6 (40%). We then tested the natural exploitation — an unbiased degree-4/6 harmonic control variate with a weight-derived basis — and report its honest failure: the estimator functions' energy at those degrees disperses across $\sim 1.8\text{e}8$ -dimensional harmonic spaces at $d=256$, so a tractable basis explains only 0.2-0.3% of per-neuron variance (measured), and the CV nets $+0.83\%$ [CI -0.6% , $+2.4\%$]. The leak is real, characterized, and — by these measurements — unscoopable by finite projection. We believe this pair (the exactness theorem + the dispersion no-go) is the sharpest available characterization of why structured spherical designs plateau on this task.

3b2. The design axis, closed from both sides: completion is exact, and worth nothing (new in v13)

This section supersedes the reachable-design part of §3b's closing claim. It revives nothing: the 129-frame completion is and remains a killed candidate (ledger `m81_full1129_pareto`, `s11_full1129_reopen_measured_breakeven`), reported here as falsification rather than as a disposition.

Our estimator integrates over 126 phased-Hadamard frames, each an orthonormal basis $H_{256} \text{diag}(\phi_s)/16$, antipodally doubled to 64,512 points on S^{255} . It is an exact spherical 2-design, and antipodally therefore a 3-design. The natural next question is whether pushing to degree-4 exactness would help. It can be answered exactly, in both directions, and the answer is instructive.

The design cannot reach degree 4 at its current size. [D] The Delsarte–Goethals–Seidel bound for an antipodal spherical 4-design in S^{255} — which is automatically a 5-design, since odd harmonics cancel pairwise — is $2 \cdot C(257, 2) = 65,792$ points. We spend 64,512. We are 1,280 points short, so no reweighting of these nodes under antipodal pair symmetry reaches degree 4.

Exactly one frame count fixes it. [D] For m mutually unbiased bases antipodally doubled, every point sees one inner product at $+1$, one at -1 , 510 at 0 , and $512(m-1)$ at $\pm 1/16$, so $\sum_y \langle x, y \rangle^4 = 2 + (m-1)/128$. A 4-design requires $3N/(d(d+2)) = m/43$. Equating and clearing $128 \cdot 43$ gives $10965 = 85m$, hence $m = 129$ and no other integer — 130 clears the counting floor and still fails. And $129 = d/2 + 1$ is the maximum number of real mutually unbiased bases in R^d when d is a power of four, which $256 = 4^4$ is. Under the Walsh doubling the complete set has d^2+2d points against a floor of d^2+d , clearing by exactly d at every level: $24/20$, $288/272$, $4224/4160$, $66048/65792$. The completed design is a near-tight antipodal 5-design, over the floor by 0.39%.

And we can say in advance what completing it is worth. [D] For a bias-free He-initialised ReLU network the rotation-averaged two-point function is exactly the iterated arc-cosine kernel $\kappa(c) =$

$(E||X||^2/d) \cdot \kappa^{32}(c)$, so the estimator's variance decomposes as $\sum_1 ||f_1||^2 A_1$ against the design defects above. That predicts $v_{126} = 2.4977e-7$ against a measured geomean of $2.6697e-7$ over sixteen fresh networks — **the variance of this estimator is predictable from first principles to 6.4%** — and it puts the degree-4 share of that variance at **0.4497%**.

So completion is worth about half a percent, against a 2.33% break-even set by the point-count cost. [O] Measurement agrees from two further directions: a point-count-matched experiment isolating degree-4 exactness at equal 66,048 points returns 0.176%, CI $[0.970, 1.028]$, $P(\text{better}) = 0.54$, and a committed degree-4 control variate returns +0.42%. Three routes, one predictive, all landing an order of magnitude below the bar.

That agreement is worth more than any of the three alone, because the analytic route explains the other two. Most of the variance is simply not where design strength lives: **86% of it sits at degrees 8 and above**, which no reachable design touches.

And degree 6, where the error actually lives, is unreachable. [D] The measured angular error sits at degree 4 (11% of the iid level) and degree 6 (40%). An antipodal 6-design needs $2^C(258, 3) = 5,658,112$ points, 87.7x our budget; any positive-weight rule needs $\dim P_3(S^{255}) = 2,861,952$, still 44.4x.

The design axis is therefore closed from both sides at once, and not for the reason one would guess. It is not that the design cannot be completed — it can be, exactly, and we did. It is that **completing it perfectly buys nothing measurable**, because the estimator's residual does not live in low-order design strength. Perfecting a design is not the same as reducing its error, and on these networks the two come apart.

Companion numbers, committed for checkability

(experiments/mub129_replication/GEGENBAUER_CHECK.json, exact rationals, even degrees): $A_4 = 7.350908201315546e-07$ at $m=126$, $2.4120167535566633e-07$ at $m=128$, exactly 0 at $m=129$; A_6 moves only $3.194089008420301e-05 \rightarrow 3.122025216144244e-05$. The design's own high-degree behaviour is the $1/N$ law — because $P_1(1/16) \rightarrow 0$, $A_1 \rightarrow 2/N$, so $A_1(129)/A_1(126) = 126/129$ to six digits for every even $1 \geq 8$. Adding points cannot beat adding points.

3c. The marginal-value map (new)

Single-component ablations of the submitted estimator on a cached-truth panel (paired seeds, CIs): the spherical design 2.02x [1.45, 2.83], antipodal pairing 1.91x [1.41, 2.56].

Erratum E8: the “exact radial conditioning 2.14x” figure was a bundled attribution and is withdrawn. The 2.141x arm ablates *two* things at once — it replaces the Kerdock frames with iid points **and** disables exact radial conditioning — so it cannot be read as the value of radial conditioning. The committed isolated factors are **frame design 2.01643x** and **residual radial improvement beyond the retained degree-2 radial control 1.06183x** (wc1_results.json, derived_isolated_ratios). The second number is small for a reason the companion P5 derives exactly: the ablated arm still retains the variance-optimal degree-2 member of the same class-A family, which alone removes 99.9861% of the radial second-moment excess. So radial conditioning is doing real but *marginal* work on top of a control that was already nearly optimal — not carrying a 2.14x pillar.

Also withdrawn with it: “three multiplicative variance pillars, each exact arithmetic or **proven locally optimal**.” Two of the three were never proven locally optimal, and the multiplicativity was never tested factorially.

Terminal folding is exactly MSE-neutral (ratio 1.00003) while saving 4.8% of billed budget; structural pruning is MSE-neutral (1.014 [0.98, 1.05]) while saving 25.1% — and the adjusted-score arithmetic makes pruning strictly optimal (removing it is 1.33x worse adjusted). The first-layer moment-tangent control measures neutral on this design (1.019 [0.97, 1.06]), consistent with three independent tests. The hosted per-network score spread (11x) is mostly rotation-draw sampling variance, with a real net-difficulty component: net-difficulty relative variance vD 0.08-0.12 (s17 σ^2 -implied; the p2-implied 0.23-0.36 is an upper sensitivity), i.e. a per-net difficulty ratio of ~ 2.7 -3.4x (experiments/s1b_dispersion_corrected/). The rotation-draw part is irreducible at fixed randomization, and the decomposition is relevant to anyone comparing per-network scores across submissions.

3d. The suite-risk decomposition, and a two-sided concentration rule (new)

Treating the SUITE as the statistical unit (the prize is one draw of a private test suite, scored by the mean) yields a decomposition we have not seen stated elsewhere in this competition. Bootstrapping synthetic suites from our measured per-network data (the 80-network tail checkpoint, and a 3-network x 16-rotation grid isolating the rotation draw; dispersion corrected in experiments/s1b_dispersion_corrected/); at $R=1$, **17-23% of across-suite score variance is net difficulty and 77-83% is rotation-draw sampling variance** (net-difficulty relative variance vD 0.08-0.12, s17 σ^2 -implied; the p2-implied 0.23-0.36 is an upper sensitivity; per-net difficulty ratio ~ 2.7 -3.4x). The corrected model brackets the independently observed 15.53x 80-net max/min spread ($P(\text{sim} \geq \text{obs}) = 0.72$ -0.86), which the original model missed entirely ($P = 0$); the spread was measured on the local synthetic 80-net tail checkpoint (m185 stage 1, a1b_tail_diagnostics.json) — the same basis as the model, and previously mislabeled “hosted” in one internal verdict, corrected here. Splitting the same billed budget across R rotations per network (equal weights) preserves the expected score while shrinking the suite-score SD by an analytic 44%/40% at $R=6$ (vD 0.081/0.122; predeclared 25% gate) — rotation-draw variance stays dominant under the correction.

The decision-theoretic consequence is two-sided, and we state it honestly: concentration removes the lucky tail as well as the unlucky one (our $P(\text{suite} < 1.6e-7)$ at $R=1$ is 9.2-10.6% under the corrected dispersion — the original model put it at 6.4%, falling to 0.01% at $R=6$; the $R>1$ arms were not re-run under the correction, but the analytic shrink above carries the same two-sided consequence). In a winner-take-all evaluation, variance is the trailing competitor’s friend and the leader’s enemy: R should be chosen by one’s expected position relative to the nearest rivals, not by per-network MSE — which R leaves untouched. To our knowledge this borrows a standard result from tournament/portfolio theory into white-box estimator design for the first time on this benchmark, and it is fully reproducible from the committed bootstraps (experiments/s1_suite_risk/, experiments/s1b_dispersion_corrected/, predeclared gates, model limitations recorded — the empirical rotation pool understates the true tail, making the shrink estimate conservative). A companion falsification closes the tempting corollary from a third independent direction: rotation-quality selection/weighting is dead not only before spending (pilot ρ -0.089, weights-only ρ 0.17) but also AFTER spending — variance statistics of the paid design sample itself correlate with realized rotation error at only ρ 0.12, because the design’s deterministic equidistribution error is invisible to iid-style sample statistics. The rotation axis is information-gated at every observation point we could construct.

The fresh-seed implication for our own entry, stated for the record: under the bracket-validated dispersion the champion's suite-score P5-P95 band is $[1.54\text{e-}7, 2.16\text{e-}7]$ on a 50-net suite and $[1.62\text{e-}7, 2.06\text{e-}7]$ on a 100-net suite, with the mean unchanged ($1.830\text{e-}7$ in every arm). **Erratum E3: those bands treat the hosted $1.830\text{e-}7$ anchor as exact, and it is not** — it is itself a 50-net measurement carrying a 9.83% standard error. Folding the anchor's own error in widens the honest 50-net band to **$[1.46\text{e-}7, 2.25\text{e-}7]$** and raises $P(\text{suite score} > 2.5\text{e-}7)$ from 0.034% to **0.57%**, a 17x increase. Still small; no longer negligible, and we would rather state the wider band. The other exposure channel in a re-run is accounting, not statistics: the residual channel is 4.5% of scored C on average (7.7% of adjusted score worst-case at C/B 0.650), conditional on $\lambda = 1\text{e}11$ remaining fixed (Rules §5.3 reserves changes).

3e. A physics derivation of the wall (new)

Four physics-framed measurements, each predeclared with kill gates, now DERIVE the empirical wall rather than merely measure it:

1. **The design's exact anatomy** (Bloch/Bragg frame). The 64,512-point phased-Hadamard design's inner-product multiset is exactly $\{0, \pm 1/16\}$; its degree-4 quadrature-error operator has exactly three eigenvalue shells, and the design's entire degree-4 advantage is the 42x suppression of a SINGLE constant mode — with the $\pm 1/16$ cancellation tuned to degree 4 only, which derives, from pure code structure, the measured 11%-vs-40% degree-4/degree-6 error split. The remaining error operator is maximally flat (participation rank $\approx N$): no Bragg modes exist for any design-side control variate to target.
2. **The residual is real-amplitude speckle** (wave-packet frame). Its angular correlation length is $36\text{-}46^\circ$ against a first-principles mean-field (arccos-kernel) prediction of 21° , monotone everywhere, with per-direction energies fitting χ^2_1 decisively (KS 0.007-0.016 at $n = 64,512$). The design's minimum angle — $\arccos(1/16) = 86.4^\circ$, fixed by the same MUB structure as (1) — sits 2x above the speckle scale: every design point is an independent draw, so no inter-point structure exists to exploit.
3. **The field is statistically homogeneous**: residual energy shows no concentration near the ReLU kink set (near/far decile ratios ≈ 1.00 against a method sensitivity of $\sim 850\text{x}$), as speckle requires.
4. **A geometric depth law**: layer-defect influence on the output field decays at a measured ~ 0.87 per layer (95x across depth 32; the first five layers carry $\sim 46\%$ of field variance, the last three 0.5%), consistent with the network contracting onto a coherence-0.975 output cone (~ 2 effective output degrees of freedom of 256) that forgets late-layer randomness. The flat mean-field prediction is rejected at 21-31x. (Both laws were subsequently half-closed from first principles: the finite-width angle-flow linearization of Jakub-Nica predicts per-layer transmission 0.890 against our measured 0.869-0.879, and the D/n kernel-fluctuation expansion predicts a correlation-length inflation of 1.58-1.87, bracketing our measured 1.70-2.20. The curve-widening shape and dispersion tightness remain honestly open; experiments/s12_finite_width_kernel/.)
5. **The floor itself** (information frame; experiments/s17_ibc_floor/). The design's exact inner-product census (1) antipodally doubles to an exact five-shell 64,512-point fingerprint — the shell counts sum to $64,512^2$ bitwise and the $\pm 1/16$ shells become exactly sign-balanced — and the equal-FLOP sampling floor σ^2/N is then read directly from the measured field variance. This is a gated lower-bound ATTEMPT, not a minimax-optimality proof. **Erratum E5: all three ratios**

below were computed at $n = 3$ and are retired. At $n = 80$ (experiments/gm_s17_reuse/) the distinct-direction ratio is **1.0044, CI [0.8450, 1.1639]** — a band straddling parity, so the champion does *not* sit below this bound; the pooled ratio is **2.01, CI [1.69, 2.33]**, which straddles the class boundary at 2.0, so the gate call is unresolved rather than obtained. The retired $n = 3$ figures were: pooled 1.79x (per net 1.63 / 2.37 / 1.37) against the per-forward floor $\sigma^2/64,512$, and 0.90x against the distinct-direction bound $\sigma^2/32,256$. Read the corrected statement as: champion and floor are **indistinguishable at the resolution we have**. With $N_{\text{eff}} \approx 38\text{k}$ effective independent draws ($\sim 60\%$ of the 64,512 forwards: an antipodal pair carries correlated even-harmonic information, so a pair of forwards is worth ~ 1.2 draws, not 2). Second signal: the empirical residual correlations at the design's own spacings are $c(0) = -1.3\text{e-}3$ and $c_{\text{even}}(1/16) = -5.5\text{e-}6$ — **small at the two shell-aggregate spacings the design realizes** (Erratum E10; earlier drafts wrote “decorrelated at every design pair,” which claims a pairwise property from a shell-aggregate measurement and is withdrawn). One disclosed formula correction, reported honestly: the predeclared four-term correlation-kernel floor formula is numerically unusable at this scale — its cross-shell coefficient is 64,000, so a sub- $1\text{e-}3$ error in $c_{\text{even}}(1/16)$ moves the predicted inflation by $O(10)$, and the naive plug-in returns 24.9, a documented artifact — so the floor is anchored on σ^2/N directly, with the kernel retained only as corroboration.

Together, **and at the level these five measurements actually earn** (Erratum E4): the estimator's error is consistent with independent χ^2_1 speckle draws whose generating structure is set by the earliest layers, sampled by a design whose single *measured* exploitable mode is already optimally suppressed. This is a **model the measurements support, not a theorem they prove** — the one-point amplitude and the correlations at realized spacings are measured; the harmonic spectrum beyond the pre-charter modes is not. Within this model, variance-per-billed-FLOP was the best lever we found and the finite-width offset in (2) is the crack we could locate. v9 said “the only lever the physics admits” and “the sole remaining crack”; both are withdrawn as overstatements of a family-local result. All five measurements are reproducible from committed artifacts (experiments/s5_kink_concentration/, s6_bragg_spectrum/, s7_speckle/, s8_layer_profile/, s17_ibc_floor/), each with predeclared gates, two-signal verification, and recorded deviations.

These measurements assemble into an achievable-envelope map $S(B)$ — the least MSE reached at FLOP budget B **among tested method classes**. It has two arms. A budget-independent **closure plateau** at $9.6\text{e-}5$: analytic degree- ≤ 2 -exact integration removes only the exactly-integrable part, and cheap first-layer covariates add $\leq 1.56\%$ out-of-sample R^2 , so more analytic effort does not lower it. A **$1/N$ sampling line** through the champion ($2.818\text{e-}7$ at C/B 0.65; $5.35\text{e-}8$ at 5.27x budget), on which the champion sits within $\sim 2\text{x}$ of the floor attempt of (5). The vertical gap between plateau and line at our budget is **340.9x** in raw final-layer MSE (384x against the $\sim 2.5\text{e-}7$ sampling point of §2; the withdrawn 524x figure mixed raw against adjusted and is not restated here — see E7, and §2 for the two admissible ratios, 340.9x raw/raw and 52.4x adjusted/adjusted). The gap is not headroom we left by sampling badly; it is the price of point-evaluation information: pay FLOPs on the sloped arm, or accept the plateau. Among the classes we tested, the only way to sit **INSIDE** the gap — low MSE at low budget — is to leave the point-evaluation oracle entirely and read the weights themselves. This is a map of tested classes, not a proof that no untested output-side method enters the gap.

The synthesis, restated at its earned level (core/GOD_NODE_SYNTHESIS_20260810.md; **substantially withdrawn — Erratum E4**). A centrality analysis of the campaign's evidence graph finds one central

node, and v9 stated it as a theorem. It is not one. The admissible statement is:

At the frozen target, the Kerdock estimator is an exact spherical 2-design whose committed degree-4 design-error operator is broadly distributed, and whose measured output residual has chi-squared-like **one-point** energy and very small correlations **at the design's realized spacings**. Multiple predeclared output-side feature families failed against it. These are strong descriptive and family-local observations. The residual harmonic spectrum beyond the pre-character measured modes, and any universal truncation or lower-bound consequence, **remain open**.

What v9 asserted and this version withdraws: that the residual is *maximum-entropy* (equipartition was never established, and the Ro computation that appeared to replace it is quarantined); that degree 4 is *the* boundary or the only suppressed degree; and that an independent-cell count N_{eff} sets a floor. A χ^2 -like one-point amplitude marginal does not determine a harmonic spectrum, and the degree- ℓ design-error operator and the residual field's harmonic energy coefficients are different mathematical objects. What survives is narrower and still useful: the champion has **no component that fits the evaluation suite**, and the N8c screen detects no material final-layer bias (E6) — which does not require the entropy claim at all, and is not the same as having no fitted constants (it has several; see the executive summary). One precision matters, and we state it carefully: the entropy is COMPUTATIONAL, not ontic. The residual is a deterministic function of weights we possess — the weights are the seed — and it **survived every sub-budget test this campaign constructed** (Erratum E10). Earlier drafts said the ledger “certifies PRNG strength.” It does not: a finite battery of failed attacks is evidence about the battery, not a certificate of pseudorandomness, and no formal indistinguishability claim is available here. What the ledger records is that the families we tried did not find exploitable structure — which leaves open both that a family we did not try would, and that the structure is there to be found.

Among the classes we tested, the 340.9x gap region was reachable only by seed-side methods — estimators using the weights at the source rather than testing the output. That is a statement about tested classes.

3f. Adversarial closure: the floor as an earned result (new)

Everything above is what survived a deliberate attempt to destroy it. Twenty adversarial agents across three fleets, all seeded, each mandated to BEAT the champion rather than to confirm it, spent a day attacking the campaign from every side we could name (`core/GEN7_ADVERSARIAL_CLOSURE_20260810.md`).

Six attack lenses against the estimator. Five returned no candidate at all, each leaving a named obstruction on the record rather than a shrug: exact-identity (all four closed-form conditional expectations are already consumed by the estimator); control-variate (the 2-design absorbs every degree- ≤ 2 statistic exactly, and degree- ≥ 4 content is $\sim 1e-5 R^2$); biased-hybrid (the baseline measures unbiased, so the MSE-optimal shrinkage weight is ~ 0 , and every realizable form came out worse, -5.7% to -38%); cost-remap (the bill is $\sim 99\%$ irreducible float32 matmul already at the floor rate, f64 share 0.033% of ops and 0.0755% of billed predict FLOPs — two different denominators, both immaterial); design-alt (the DGS bound needs $N \geq 33,152$ for a 4-design and $\sim 44x$ the rows for degree-6 nulling, and Var-C is invariant on the flat speckle of $3e(2)$).

The one candidate our own theory could not exclude. The sixth lens found it: seed-side rotation CONSTRUCTION — replace the grader-seed Haar input rotation with the deterministic V from the first

layer's SVD ($W_0 = U S V^T$), coupling the randomization to the weights we already possess. This is the lane the speckle account of §3e leaves formally open, and the attacker predicted a win from a 24-rotation oracle spread of $5.8e-8$ to $6.8e-7$ with the champion sitting at the mean. Measured on the committed public 0-99 net basis at seed 0 under common random numbers, twice-run bit-identical (noise floor exactly 0.0), it is a clean null: paired $t = +0.19$ against a $t \geq 3$ kill gate, the variant better on exactly **50 of 100** nets, a bootstrap CI on the mean delta of **$[-3.19e-8, +2.61e-8]$** symmetric about zero, and a drop-top-5 recomputation that flips the sign — bought with $+3.6e8$ billed FLOPs (C/B $+0.0014$) and no compensating MSE reduction. The mechanism of the null is that V is marginally Haar: coupling it to the weights' singular basis buys no alignment with the fixed cubature frame (ledger 242, killed).

Seven kill families re-litigated under changed premises: 7/7 held, several strengthened. The fidelity family formally retired the dtype-repricing escape (the honest f64 share is 0.0755% of billed predict, so a complete recast is capped at 59,656,312 FLOPs — immaterial however f64 is priced; v9 filed 0.00% here from a void detector, and the conclusion is unchanged, see Erratum E1); the closure family's four insertion points were re-killed under maximally favourable cost assumptions; and the dispersion family named the one un-probed crack left in the whole record — non-smooth cell-membership covariates, inside a profitability window of $R^2 \in [2.63e-5, 1e-4]$.

S18 seals that crack the same day (experiments/s18_cell_membership_probe/, ledger 241).

Arrangement-combinatorial features — cell-membership indicators at $k = 16/64/256$ and Hamming distance to the modal sign pattern — were gated as incremental regressors beyond the degree- ≤ 2 basis on the S5/S15 trio, using S15's exact split. Every gated set lands below the predeclared per-coefficient fitting-noise cost of **$2.63e-5$** on all three nets (best gated value **$2.371e-5$** , 0.9x the bar and inside its own permutation null of $|5.3e-5|$), and none reaches the $1e-4$ SIGNAL bar on any net (0 of 3, versus 2+ required). The mechanism is structural and identical on all three nets: all **64,512** design directions occupy **64,512 DISTINCT** first-layer activation cells — every cell a singleton, maximum cell count 1 — so at design spacing cell identity is a per-point unique label, which can memorize one training residual and can never activate on the held-out half. Only cell aggregates can generalize, and those measure at fitting noise. Instrument checks: S15's Base-B out-of-sample R^2 reproduced exactly ($0.3609 / 0.4037 / 0.4385$), an injected $1e-3$ signal recovered at $1.53e-3 / 0.89e-3 / 0.71e-3$, and an independently recomputed first-layer pass reproducing the cached S5 arrays bit-exactly (max absolute difference 0.0 on all three nets).

What this establishes, and what it does not. It does not upgrade §3e(5): the floor there remains a gated lower-bound attempt, not a minimax-optimality proof, and this section adds no theorem to it. What it does establish is the status of the claim itself. The champion's position is not an assumption and not a self-assessment: it is the statement left standing after twenty agents whose explicit job was to break it failed against seeded, deterministic, predeclared gates — including the single seed-side construction our own theory could not rule out in advance — while the same campaign turned on our own record and forced four same-day corrections, two of which are visible in this document (§3e(5)'s floor language de-escalated to S17's own "attempt", §3c/§3d's dispersion re-measured and bracket-validated). The question we leave open, and state as open: whether NON-rotation seed-side structure is extractable at all. Earlier drafts said "the rotation lane closes at the point-evaluation level." **Withdrawn (E10)**: one seed-side rotation construction was built and measured a clean null (paired $t = +0.19$), and a lane is not closed by a single null. The accurate statement is that the one rotation-side construction we

built did not work, and the deeper lane is untested here. Every number in this section is reproducible from committed artifacts.

4. Methodological notes: calibrate your suite, and re-measure your graveyard

Local development suites are not the grader's suite. We measured ours by running a budget-matched Monte-Carlo reference on both. **Calibrate on the median, not the mean** — and this correction is one we had to apply to ourselves (Erratum E2). Our filed v9 reported local $1.069\text{e-}6$ against the grader's published $6.47\text{e-}7$, a ratio of 1.65, and advised the field to apply that factor. Recomputing the same committed 22-net panel by median gives $6.47355\text{e-}7$ against the grader's $6.470\text{e-}7$ — **a 0.05% match**. The 1.65 was entirely an artifact of the mean, which the per-net score distribution's ~11x right tail dominates. There was no suite-difficulty shift to apply, and the recommendation as filed was wrong.

The methodological point survives in stronger form. Local development suites are still not the grader's suite, and a one-run calibration is still worth doing before any cross-suite comparison — including comparisons against published competitor numbers. But run it on the median. On a heavy-tailed per-net distribution the sample mean is a statistic about the tail, not about the suite, and comparing two means across suites measures which one happened to draw a worse outlier.

Falsification hygiene: the same standard applies to your dead entries. A record-level sweep of our full ledger (~307 mechanism and uncertainty records, read one at a time; `core/GRAVEYARD_MINE_20260810.md`) found dispositions carried by wording rather than by evidence — three “kills” had never been run at all — so we ran 16 falsifiers on the strongest candidates and adjudicated all 16 (`core/GRAVEYARD_RUN_RESULTS_20260810.md`, with a predeclaration, verdict and results file per item under `experiments/gm_*/`). Ten historical dispositions converted from assumption to measurement and stand as confirmed kills; four revived as SCREENED Phase-2 proposals, which are proposals and not promotions; one is blocked for escalation and one is recorded as an honest inconclusive hold. The one revival that is a result in its own right is a dtype convention: the float32-parity precondition behind the rank-one square's cost had been assumed rather than measured, and discharging it (worst predeclared identity error $2.81\text{e-}6$ against a $1\text{e-}5$ gate, with an exact-rational reference at $2.02\text{e-}16$, an alternate-association cross-check at $1.98\text{e-}15$, and a bitwise re-run in fresh interpreters) reprices the affected analytic bills about 2x down — screened only, measured at width 256, not reusable at larger widths without re-measurement, residual risks named on the record, nothing deployed and no Phase-1 score headroom claimed anywhere in the sweep.

4b. LLM involvement, stated in full, and what we did about it

Official Rules v12 §5.7 *encourages* prize-eligible solutions to disclose “what LLM tools — if any — were materially used in generating or substantially modifying the Solution code, and the extent of that use,” and §6 lists “the clarity and accuracy of the technical writeup” among the judging criteria. We disclose well past what is asked, because the honest answer here is unusual enough that a disclaimer would misrepresent it.

Erratum E11. Earlier drafts, including the version filed 2026-08-10, opened this section by saying the guidelines “require full transparency about LLM involvement and warn that unhedged dubious claims reduce credibility.” **Both halves were wrong.** Disclosure is encouraged, not required; and the second clause is not organizer text at all — a full-text search of the challenge forum for *unhedged*, *dubious* and

credibility returns zero results, and the phrase appears nowhere in Rules v12. We had been citing, as the governing standard for this entire document, a sentence the organizers never wrote. It is the sharpest instance we have of the failure mode this section exists to guard against, and we found it the only way such things are ever found: by going and reading the primary source instead of our own summary of it.

Disclosure. This campaign was conducted end-to-end by large language models operating as agents, under human direction. The estimator, the experiments, the ledger, this document and the six companion papers were all written by LLMs. Nothing here is human-authored code with LLM assistance; it is the other way round. We think that is exactly the situation the guidelines are worried about, and the rest of this section is what we built to make it auditable.

The evidence-tag discipline. Every load-bearing claim in this document and in all six papers carries one of four labels introduced in companion P1 (front matter, lines 38-42 — **not** the corpus README, which does not contain the scheme): **[O]** observed (a run in this corpus produced it), **[D]** derived (follows by steps shown inline), **[R]** reported (a committed artifact says so, not re-derived here), **[A]** assumed (a stated modelling choice). A fifth, **[GAP]** — a known hole with the check that would close it — is used in this document and in P5 and P6 but is **not** part of P1’s scheme; *erratum E13*, since earlier drafts attributed all five to P1 and cited a “P1 §0” that does not exist. The rule is that no claim may be stated above the level its artifact earned. The tags exist because a fluent model produces plausible text whether or not the underlying claim is true, so fluency has to be decoupled from confidence mechanically rather than by intention.

The adversarial protocol. Two LLM agents from different model families worked the same corpus under a sealed evidence charter: each committed proposals by SHA-256 before revealing bytes, each audited the other’s mathematics, and neither could admit a measurement as evidence unless it was disclosed as in flight before it ran. A third mechanism — a twelve-agent refinement pass over the six papers, with adversarial verifiers instructed to *refute* rather than confirm — ran independently of both.

What that machinery found, in one day, in our own flagship claims. We report this because it is the strongest evidence we can offer that the labels above are load-bearing rather than decorative:

- **A false uniqueness theorem.** Our proof that uniform weights are the unique constrained minimiser at every even degree is false at degree 2: the design is an exact 2-design there, so the kernel matrix has a 126-dimensional null space and every per-frame reweighting is free. Found by three independent routes within hours — a two-frame witness, the full kernel dimension, and an exact rational recomputation. The global-minimiser half survives and is what §0 and the papers now claim.
- **An unproved exhaustiveness step.** The “no third class” clause of our divergence-form dichotomy was asserted under hypotheses too weak to carry it; an admissible field of the form $s \cdot x / |x|^{(d+1)}$ produces a third singular source at the origin. The definition has been strengthened and the branches are unaffected.
- **A control with no power, presented as a control.** A permutation null we cited as evidence is provably incapable of returning anything but the value it returned — our own corollary forces it. Its residual was *smaller* than the real arm’s, which is the signature of re-measuring an identity.
- **An inadmissible measurement supporting a headline.** The computation underpinning our central spectral claim was executed after the evidence charter without disclosure. It is now

quarantined — retained and struck rather than deleted — and the claim it supported is **open**.
 Withdrawing it does not revive the account it replaced, which was never established either.

Several published figures were also wrong at the third significant digit or in their normalisation, and are corrected in §0 and in the papers with the arithmetic shown.

The asymmetry we want on the record. Public scrutiny of this work — a forum sweep across every competitor write-up and organizer thread we could find — produced **no** defect in our record. Every defect above was found by our own machinery pointed at ourselves. We do not offer that as proof of correctness. We offer it as the reason to believe the remaining claims are labelled honestly: the same process that produced them is demonstrably willing to destroy them.

5. Compute transparency

Our submissions bill at $\sim 7\text{--}8e10$ analytical FLOPs per second of wall time; the graded submission runs at C/B 0.650 (1.768e11 scored FLOPs). Our estimator's residual (uninstrumented) exposure is 4.5% of scored C on average and 7.7% of adjusted score in the worst case at that C/B, conditional on $\lambda = 1e11$ remaining fixed; any tightening of residual accounting under §5.3 (which the Rules reserve) would cost us at most that worst-case 7.7% and nothing else.

6. Reproducibility

The estimator source, predeclarations, kill gates, adversarial audits, frozen manifests, and the 267-record fold ledger are at

```
github.com/gmrnk/recursive-estimator-folding
/tree/f225be4e4e4872dc2bef06711525cf00e73a332b/corpus/whestbench
```

Read that path, not the repository root. The default branch is a 2026-08-06 snapshot and does not contain the papers, this write-up, the ablation results, or the current ledger; the ledger it does carry has 43 records rather than 267, which would contradict this section. The campaign branch is agent/compression-survivor-corpus, and the filing pins an exact commit so the citation cannot drift. The pinned commit is the pushed branch tip of 2026-08-17, verified against origin before filing. **The repository was made public on 2026-08-17, immediately before this filing**, so the citation resolves for any reader without an access request.

The repository contains no challenge data, no private truth, no scorer, no credentials, and not the #326094 submission archive. Negative results are retained in full rather than pruned, which is what makes the falsification ledger checkable rather than merely asserted.

Erratum E12. Earlier drafts, including the version filed 2026-08-10, cited the bare repository URL and stated that all artifacts and the 267-record ledger were there. On the default branch they are not. A reader following that citation would have found a stale tree whose ledger disagrees with this document by 224 records. The artifacts were always present on the campaign branch; the citation pointed at the wrong place. A second, related defect is corrected in this draft: earlier drafts described the corpus as public while the repository was in fact private. It was made public on 2026-08-17, immediately before this filing, so the description and the state now agree.